

# Competitive Bank Liability Design

By SAKI BIGIO, PIERRE-OLIVIER WEILL, AND DIEGO ZÚÑIGA\*

## I. Introduction

At its core, banking is about the swap of assets for liabilities. When it purchases a security or makes a loan to a business owner, a bank acquires an asset. In exchange, the seller of the asset or the borrower receives a deposit, a bank liability. The deposit is liquid because it circulates: for example, the business owner can use it to pay their workers, who in turn can use it to make further purchases, and so on. Is the liability-creation process purely accounting, or does it promote welfare? How do banks create circulating liabilities, and should these liabilities have stable payoffs? Views on these classical questions shape debates on monetary policy and financial regulation.

To study these questions, we follow the approach of [Bigio, Weill and Zúñiga \(2025\)](#): we reformulate the banking theory of [Gorton and Pennacchi \(1990\)](#) in a monetary economy, where securities are explicitly used as means of payment, and banks design their liabilities optimally. We consider a three-period, two-state economy with multiple assets in positive supply. Assets differ in risk exposure: consider fixed-income securities, mortgages, consumer or business loans, and equities. There is a continuum of identical producers endowed with all of the economy's assets and a continuum of workers who supply labor. Producers and workers are matched bilaterally and anonymously to trade labor. As in the literature following [Lagos and Wright \(2005\)](#), both producers and workers lack commitment. This precludes the use of credit and implies that producers must use assets to pay workers. As in [Rocheteau \(2011\)](#), we assume that producers receive private information about their asset payoffs before matching with workers. This creates an asymmetric information problem and implies that assets with high risk exposure are illiquid and cannot circulate as a means of payment. Assets with moderate risk, on the other hand, are liquid: they always circulate as means of payment.

In this context, we introduce agents who can com-

mit, bankers. Bankers compete to swap producers' assets for liabilities before the arrival of private information. As in the security design literature (for example [DeMarzo and Duffie, 1999](#); [Biais and Martiotti, 2005](#), among many others), bankers choose the state contingent payoffs of the liabilities they issue.

In equilibrium, bankers engage in liquidity transformation, creating value in two ways. First, bankers can bundle the producers' assets into a portfolio that may, on its own, circulate as a means of payment. Second, if the bundle remains illiquid, bankers can tranche it into several securities, some illiquid and others liquid. Correspondingly, we find that bankers issue two types of liabilities: a liquid liability, resembling a deposit security, that circulates as a means of payment, and an illiquid liability, resembling bank equity, that does not circulate because of asymmetric information.

We emphasize three key implications of this banking theory. First, bankers only create value by holding illiquid assets and swapping them for liquid liabilities. Hence, a narrow bank would not create any value; it would just swap liquid assets for equally liquid liabilities.

Second, although a bank cannot create value by holding only liquid assets, it can create value by bundling liquid and illiquid assets. Bundling assets increases the total amount of securities that can circulate as a means of payment. Thus, liquid and illiquid assets are complementary inputs in liquidity transformation.

Third a commonly held view is that means of payments, such as money, bank deposits or cryptocurrencies, should be riskless. Contrary to this view, we find that making deposits moderately risky does not reduce their liquidity and maximize producers' total payment capacity.

Of course, these implications are specific to our model. In reality, banks may create value in other ways and may face incentive problems that can be mitigated by promoting narrow banking or stable deposits. Our main observation is that such policies involve a trade-off, because they reduce the value banks can create by engaging in liquidity transformation.

\* Bigio: Department of Economics, University of California Los Angeles and NBER, ([sbigio@econ.ucla.edu](mailto:sbigio@econ.ucla.edu)); Weill: Department of Economics, University of California Los Angeles, NBER, and CEPR ([poweill@econ.ucla.edu](mailto:poweill@econ.ucla.edu)); Zúñiga: Bank of Canada ([diego.zuniga.seminario@gmail.com](mailto:diego.zuniga.seminario@gmail.com)).

## II. The Economic Environment

There are three dates,  $t \in \{0, 1, 2\}$  and two possible states,  $\omega \in \{\ell, h\}$ , with corresponding probabilities  $\pi_\omega$ . There is a single consumption good and a continuum of risky assets in positive supply, which we call trees. A tree entitles its owner to a positive state contingent payoff at time  $t = 2$ , represented by the vector  $x = (x_\ell \ x_h) \in \mathbb{R}_+^2$ . Trees represent broad asset classes, such as bonds or equity. Trees are divisible. Property rights to tree shares can be traded.

The economy is populated by three types of agents, with a measure one continuum of each: producers, workers, and bankers. All agents are risk-neutral and enjoy consumption at  $t = 2$ . Producers are endowed with all the trees at  $t = 0$ .

At  $t = 1$ , each producer operates a linear production technology, whereby  $q$  labor units yield  $\rho q$  units of the consumption good at  $t = 2$ , regardless of the state. Workers supply labor at a constant marginal disutility cost normalized to one unit of the consumption good. We make two key assumptions. First, producers have private information about the realization of  $\omega$  at  $t = 1$ . Second, producers and workers lack commitment, precluding the use of credit in bilateral trade. Hence, as in the monetary literature, producers must use securities as a means of payment to purchase labor. Crucially, since producers have private information about the state dictating asset payoffs, using securities to pay for labor is plagued by asymmetric information.

Bankers can neither access the production technology nor supply labor. Instead, unlike producers and workers, bankers can commit to make  $t = 2$  state-contingent payments or, equivalently, have the financial skills to structure payoffs and the legal skills to write enforceable contracts. As a result, bankers are in a unique position to issue liabilities that can be used as a means of payment. The timing is as follows:

- $t = 0$ : competitive liability design. Bankers compete to swap the producers' tree portfolios for bank-designed liabilities.
- $t = 1$ : decentralized exchange. Producers learn the aggregate state,  $\omega$ , and are bilaterally and anonymously matched with workers. Labor is traded for securities under asymmetric information.
- $t = 2$ : payoff, settlement, and consumption. Output is produced, all trees pay off, contracts

with bankers are settled, and all agents consume.

In the next two sections, we solve for an equilibrium by backward induction. Section III studies decentralized exchange at  $t = 1$ , and Section IV analyzes the banks' competitive liability design at  $t = 0$ .

## III. Decentralized exchange

Consider a producer who seeks to hire labor from workers using an arbitrary state-contingent security with payoff  $x = (x_\ell \ x_h)$ . This security could be a tree in the endowment of the producer, or a liability designed by a bank.

In an efficient trade, the workers would supply  $q = \mathbb{E}[x]$  units of labor in exchange for the security, and produce  $\rho \mathbb{E}[x]$  consumption goods. However, since the producer knows the state  $\omega$  and the workers do not, trade occurs under asymmetric information possibly precluding efficient trades.

As is well known, the exact outcome of the trade under asymmetric information depends on the trading protocol. We adopt the trading game of [Attar, Mariotti and Salanié \(2011\)](#) in which workers compete for non-exclusive trade with producers. In [Bigio, Weill and Zúñiga \(2025\)](#), we show that the outcome depends on two coefficients:

$$\psi_\ell \equiv \frac{1 - \rho \pi_\ell}{\rho \pi_h} < 1 < \psi_h \equiv \frac{\rho \pi_\ell}{1 - \rho \pi_h},$$

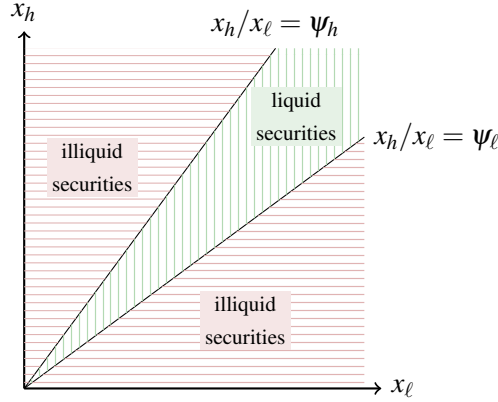
where we maintain the assumption that  $\rho \pi_\omega < 1$  for all  $\omega$ .

When the security has moderate risk, i.e., its payoff does not vary much across states,  $x_h/x_\ell \in [\psi_\ell, \psi_h]$ , the security is liquid. In turn, when the security is liquid. That is, in equilibrium, trade is efficient, as the security trades at the pooling price  $q = \mathbb{E}[x]$ . The ex-ante value of the producer for a liquid security is equal to

$$\mathbb{U}[x] = \rho \mathbb{E}[x],$$

since the worker is supplying  $\mathbb{E}[x]$  unit of labor.

If the security is sufficiently risky, in that its payoff varies substantially across states, then it is illiquid. The pooling price can no longer be the basis of an equilibrium: the pooling price is too low when the security payoff is largest, so the producer prefers to hold on to the security. For example, suppose that  $x_h > \psi_h x_\ell$ . Then, in the low state, the security trades for  $q = x_\ell$  units of labor, leading to a production of



**Figure 1. :** The cone with vertical hatching represents liquid securities. The two cones with horizontal hatching represent illiquid securities.

$\rho x_\ell$ . In the high state, the security does not trade, and the producer enjoys utility  $x_h$ . Thus, for illiquid securities, the ex-ante value of the producer is equal to:

$$\mathbb{U}[x] = \begin{cases} \rho \pi_\ell x_\ell + \pi_h x_h & \text{if } x_h/x_\ell > \psi_h \\ \pi_\ell x_\ell + \rho \pi_h x_h & \text{if } x_h/x_\ell < \psi_\ell. \end{cases}$$

In Figure 1, the set of liquid securities is represented by the cone with vertical hatching, and the set of illiquid securities by the two cones with horizontal hatching. Notice that the function  $\mathbb{U}[x]$  is discontinuous: it jumps down when moving from the set of liquid to the set of illiquid securities. This discontinuity is the symptom of illiquidity: it arises because, when the security becomes illiquid, trade breaks down in one of the two states.

#### IV. Competitive liability design

At  $t = 0$ , producers are endowed with the trees. We represent this endowment with a positive and finite measure  $\mu_0$  over  $\mathbb{R}_+^2$ . Thus, we do not impose any restriction on the endowment: it may consist of any combination of liquid and illiquid trees and may comprise continuous or discrete measures or a mixture of both.

Bankers offer to swap these trees for portfolios of bank-designed liabilities,  $\mu_1$ , also represented with a positive and finite measure over  $\mathbb{R}_+^2$ . Because producers are identical, hold the entire tree endowment, and because bankers are competitive with linearly homogeneous budget sets, it follows that any equilibrium portfolio of bank liabilities must maximize the producers' utility, subject to a zero expected profit condition. This implies that bankers' consumption

must be zero in both states, since it cannot be negative, and that the aggregate payoff of the portfolio  $\mu_1$  they offer must be equal to the aggregate payoff of the producers' tree endowment.

In sum, an equilibrium portfolio of bank-designed liabilities solves the following problem:

$$(1) \quad \max \int \mathbb{U}[x] d\mu_1(x),$$

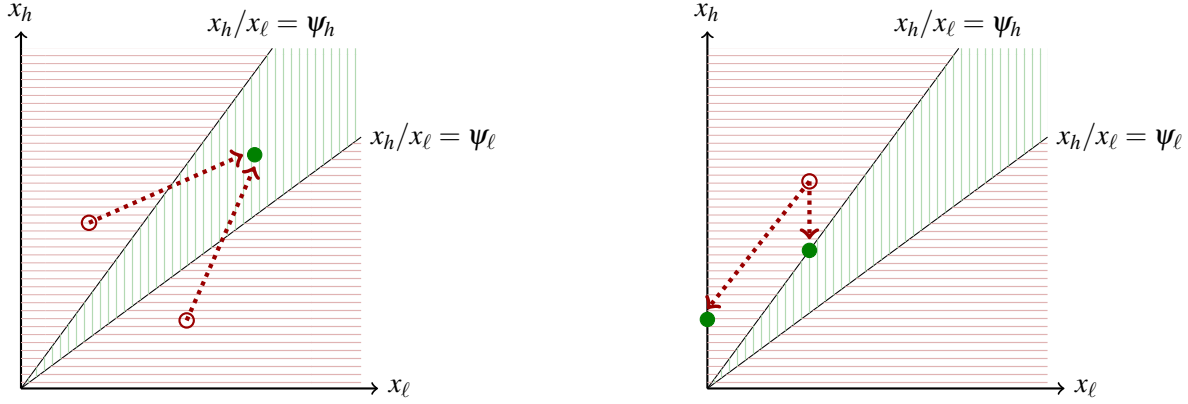
with respect to a portfolio of bank liabilities,  $\mu_1$ , and subject to

$$(2) \quad \int x_\omega d\mu_1(x) \leq \int x_\omega d\mu_0(x),$$

for each  $\omega \in \{\ell, h\}$ . The following proposition, proved in the appendix, gives the solution:

**PROPOSITION 1:** *Let  $X_\omega = \int x_\omega d\mu_0(x)$  denote the aggregate payoff of the tree endowment in state  $\omega$ . Then, equilibrium portfolios of bank liabilities are as follows:*

- *Liquid aggregate tree,  $X_h/X_\ell \in [\psi_\ell, \psi_h]$ : any feasible portfolio of liquid liabilities can be the basis of an equilibrium;*
- *Illiquid aggregate tree, case 1,  $X_h/X_\ell > \psi_h$ : the equilibrium portfolio is composed of one liquid liability with payoff  $(X_\ell - \psi_h X_\ell)$  and one illiquid liability with payoff  $(0 - X_h - \psi_h X_\ell)$ ;*
- *Illiquid aggregate tree, case 2,  $X_h/X_\ell < \psi_\ell$ : the equilibrium portfolio is composed of one liquid liability with payoff  $(X_h/\psi_\ell - X_h)$ , and one illiquid liability with payoff  $(X_\ell - X_h/\psi_\ell - 0)$*



**Figure 2 :** The left panel shows how bankers create value by bundling. The right panel shows how they create value by tranching.

The Proposition shows that bankers engage in liquidity transformation, creating value in two ways: by bundling and by tranching securities.

If, as in the first bullet point, the aggregate tree payoff is liquid, then bankers create value by bundling illiquid trees to make them liquid. This is illustrated in the left panel of Figure 2. The two empty circles represent illiquid trees from a producer's endowment. The filled circle represents a newly designed bank liability obtained by bundling these two illiquid trees, forming a single security whose payoffs are the sum of the payoffs of both trees in each state. Hence, bundling conserves the aggregate payoff of the two illiquid trees but makes the resulting liability liquid as it becomes sufficiently safe once payoffs are pooled.

If, as in the second and third bullet points, the aggregate tree payoff is illiquid, then liquidity cannot be generated by bundling: some of the liabilities designed by the bankers must necessarily remain illiquid. In that case, bankers create further value by tranching the aggregate portfolio of illiquid trees into two liabilities: an illiquid liability and a liquid liability. This is illustrated on the right-panel of Figure 2. Bankers maximize the amount of efficient trade by maximizing the expected payoff of the liquid liability and, vice versa, minimizing the expected payoff of the illiquid liability. For instance, if  $X_h > \psi_h X_\ell$ , then the newly designed liquid liability has payoff  $X_\ell$  in the low state, and the largest possible payoff consistent with liquidity in the high state, that is,  $\psi_h X_\ell$ . The newly designed illiquid liability has the residual payoff: zero in the low state, and  $X_h - \psi_h X_\ell$  in the high state.

Proposition 1 is related to [Farhi and Tirole \(2015\)](#)

who study bilateral trade under asymmetric information with two assets, one perfectly safe and liquid, and the other maximally risky (with a payoff of zero in one of the two states) and illiquid. They study when bundling dominates tranching, and vice versa. In their setting, bundling means trading with the portfolio of two assets, and tranching means trading with each asset separately. Our approach is different because bankers are not constrained to make a binary choice between bundling or tranching. In fact, we find that they find it optimal to do both: they bundle all of the producers' assets on the left side of their balance sheet, and tranche the bundle optimally to maximize the value created in bilateral trades.

**Interpretation and Implementation.** It is natural to interpret the liquid liability as a bank deposit, as it is a means of payment: at time  $t = 1$ , producers transfer the liability to the workers to hire  $\mathbb{E}[x]$  labor units. On the other hand, the illiquid liability is not used at all as a means of payment: the producer holds on to it because asymmetric information precludes trade with this type of security. Taken together, given its illiquidity and riskiness, it is natural to interpret this security as bank equity. [Bigio, Weill and Zúñiga \(2025\)](#) go further and present an implementation of the equilibrium outcome in this economy with collateralized loans: under this interpretation, banks do not explicitly swap trees. Instead, they make loans collateralized by trees and create deposits that they credit to producers who use them to pay the workers they hire.

**Regulatory implications.** The Proposition not only has implications for the liabilities that bankers issue, but also for the assets they acquire. In particular, acquiring only illiquid trees is suboptimal.

For a simple example, suppose that producers are endowed with two types of trees: one liquid tree with payoff  $(1 - \psi_h - \varepsilon)$ , and one illiquid tree with payoff  $(1 - \psi_h + \varepsilon)$ , for some small  $\varepsilon$ . A feasible liability design for the bankers would be to let the producer trade with the liquid tree, and tranche the illiquid tree into two securities,  $(1 - \psi_h)$  and  $(0 - \varepsilon)$ . However, this is clearly suboptimal: bundling all the securities would instead enhance the liquidity of the illiquid tree's payoffs. This observation demonstrates that liquid and illiquid trees are complementary inputs for liquidity transformation.

In this environment, two traditional policy recommendations aimed at promoting stability in the banking system would clearly destroy value. Consider first the “narrow banking” recommendation, which stipulates that on the asset side of their balance sheet, banks must only hold safe and liquid assets. In this environment, such a narrow bank would swap liquid assets by equally liquid liabilities without creating any value. Second, consider the policy recommendation to guarantee the safety of the liability side of banks' balance sheets, i.e., that deposits should be riskless. In this environment, this policy would clearly destroy some value by leaving payment capacity on the table. Maximizing producers' payment capacity requires deposits to have some moderate risk, albeit not so much as to preclude trading.

Of course, these implications do not imply that in reality there are no other reasons to promote narrow banking or stable deposits. It simply means that there is a trade-off, since these policy recommendations reduce the value banks create by engaging in liquidity transformation.

## V. Conclusion

In Bigio, Weill and Zúñiga (2025), we first formulate the model summarized in this paper, but the setting in that paper is more general along several non-trivial and important dimensions. First, we consider producers with heterogeneous endowments and thus different incentives to swap their trees for bank liabilities. We show that, in equilibrium, producers swap all their trees for portfolios of bank liabilities and that, again, bank liabilities consist of a single deposit and, if necessary, an equity claim. Second, we allow banks to start with some initial endowment. We show that, as banks start with lower initial wealth, for example, because of poor returns on their legacy assets, the optimal design of banks' deposit liabilities becomes riskier. Third, we explicitly consider

that producers and bankers can trade securities in a time-zero market. This allows us to derive asset-pricing implications, in particular, articulating which assets command a liquidity premium. Finally, we extend the analysis to consider producers that can trade with an asset (outside money) that is more recognizable than bank deposits. This extension delivers predictions regarding the holdings of outside money by banks, to enhance liquidity transformation, and the holdings of outside money by producers, to trade directly. In particular, it shows that worsening adverse selection increases banks' holdings of outside money, whereas the worsening of the recognizability of bank deposits decreases them.

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## PROOF OF PROPOSITION 1

Consider the first case in which the aggregate tree payoff is liquid,  $X_h/X_\ell \in [\psi_\ell, \psi_h]$ . Then the result is obvious because an upper bound on (1) is obtained by noting that  $\mathbb{U}(x) \leq \rho(\pi_\ell x_\ell + \pi_h x_h)$ . This upper bound is achieved by any portfolio of liquid liabilities, for instance, the liquid liability obtained by bundling all trees in the producers' endowment.

Next, consider the second case in which the aggregate tree payoff is illiquid. Then, one sees that it is strictly suboptimal for bankers to issue interior illiquid liabilities.

Consider for example liabilities with payoff  $x_h > \psi_h x_\ell > 0$  (the  $\psi_\ell x_\ell > x_h > 0$  case is symmetric). Indeed, consider any feasible portfolio and its restriction,  $\nu$ , to the set of securities such that  $x_h > \psi_h x_\ell > 0$  and suppose that  $\nu$  is strictly positive. Now replace  $\nu$  by a portfolio  $\nu'$  with two atoms. The first atom, at the illiquid security  $(0 \ 1)$ , has measure  $\int (x_h - \psi_h x_\ell) d\nu(x)$ . The second atom, at the liquid security  $(1 \ \psi_h)$ , has measure  $\int x_\ell d\nu(x)$ . By construction, the new portfolio,  $\nu'$ , conserves aggregate payoff so that (2) is satisfied for all  $\omega$ . Moreover, it increases utility strictly since

$$\int \mathbb{U}(x) d\nu'(x) - \int \mathbb{U}(x) d\nu(x) = (\rho - 1)\pi_h \psi_h \int x_\ell d\nu(x) > 0.$$

Next, we show that it is strictly suboptimal to issue, at the same time, liabilities with payoff  $x_\ell = 0$  and liabilities with payoff  $x_h = 0$ . Indeed, consider any feasible portfolio and its restriction  $\nu$  to the set of such liabilities. Suppose that

$$\nu(\{x : x_\ell = 0\}) \times \nu(\{x : x_h = 0\}) > 0.$$

Now replace  $\nu$  by the measure  $\nu'$  with two atoms. The first atom is a riskless security with payoff  $\min\{\int x_\ell d\nu(x), \int x_h d\nu(x)\}$  in both states. The second atom is the residual security. For instance, if  $\int x_h d\nu(x) \geq \int x_\ell d\nu(x)$ , then the residual security has a payoff of zero in the low state, and a payoff of  $\int x_h d\nu(x) - \int x_\ell d\nu(x)$  in the high state. By construction, the new portfolio,  $\nu'$ , conserves aggregate payoff so that (2) holds in both states. Moreover, it increases utility strictly since

$$\int \mathbb{U}(x) d\nu'(x) - \int \mathbb{U}(x) d\nu(x) = (\rho - 1) \min\left\{\int x_\ell d\nu(x), \int x_h d\nu(x)\right\}.$$

Now turning to the optimal portfolio, assume that  $X_h > \psi_h X_\ell$ . Then, any feasible portfolio must comprise a positive measure of illiquid liabilities, i.e. securities with payoff  $x_h > \psi_h x_\ell$ . Our preliminary results then imply that an optimal portfolio of bank liabilities can contain only one type of illiquid liability: the extreme illiquid security with zero payoff in the low state, and strictly positive payoff in the high state. Let us denote by  $Y_h$  the aggregate payoff of this illiquid liability, and by  $(X_\ell \ X_h - Y_h)$  the aggregate payoff of the liquid liabilities. The optimization problem thus reduces to maximizing:

$$\rho[\pi_\ell X_\ell + \pi_h (X_h - Y_h)] + \pi_h Y_h,$$

subject to  $Y_h \leq X_h$  and  $X_h - Y_h \leq \psi_h X_\ell$ . The first constraint is the feasibility constraint in the high state, and the second constraint ensures that the residual payoff is liquid. Since  $\rho > 1$  it follows that the optimum is attained by choosing the smallest feasible  $Y_h$ , that is,  $Y_h = X_h - \psi_h X_\ell$ , and we are done.